

Que 5 →

- Clone <https://github.com/zembutsu/docker-sample-nginx.git> on your local system.
- Initialize one repo on your local system named Cloudethix-nginx-yourname and then copy all the content from cloned repo to your local repo.
- Update the index.html and add the name CloudEthiX along with Hostname. Commit the change to your local repository. Then create and push the code from the local repository to your remote repository named Cloudethix-nginx-yourname under master branch.
- Once code is pushed, create a Docker hub public repository named Cloudethix-nginx-yourname.
- Then build the docker image out of the dockerfile. Once an image is built, tag the image and push that image to the docker hub repo you created.
- Create a deployment.yaml file with 2 replications in cloudethix_yourname Namespace and use the container image that you have pushed to your docker hub repo. Once a file is created, run the deployment in k8s cluster.
- Please prepare a document along with the snap and share it across

```
achal@LAPTOP-# ls
Dockerfile LICENSE README.md default.conf deployment.yml index.html
achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ cat Dockerfile index.html
deployment.yml
```

```
FROM docker.io/library/nginx:alpine
```

```
COPY default.conf /etc/nginx/conf.d/
COPY index.html /usr/share/nginx/html/
```

```
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX Nginx</title>
</head>
<body>
  <h1>CloudEthiX</h1>
  <h2>Hostname: nginx</h2>
</body>
</html>
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloudethix-nginx
  namespace: cloudethix-achal

spec:
  replicas: 2
```

```
selector:
  matchLabels:
    app: cloudethix-nginx

template:
  metadata:
    labels:
      app: cloudethix-nginx

  spec:
    containers:
      - name: nginx
        image: docker.io/achalpadol/cloudethix-nginx-achal:v1
        ports:
          - containerPort: 80
```

```
achal@LAPTOP-HPUOTJF1:~$ ls
01_pod_nginx.yaml  configmap.yaml  docker-sample-nginx
```

```
achal@LAPTOP-HPUOTJF1:~/docker-sample-nginx$ cd ~
```

```
achal@LAPTOP-HPUOTJF1:~$ mkdir Cloudethix-nginx-achal
```

```
achal@LAPTOP-HPUOTJF1:~$ cd Cloudethix-nginx-achal/
```

```
achal@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
```

```
achal@LAPTOP-# git branch -M master
```

```
achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ cp -r ~/docker-sample-nginx/.
.
```

```
achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ ls
Dockerfile  LICENSE  README.md  default.conf  index.html
```

```
achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX Nginx</title>
</head>
<body>
  <h1>CloudEthiX</h1>
  <h2>Hostname: nginx</h2>
</body>
</html>
```

```
achal@LAPTOP-# git log --oneline
2f44351 (HEAD -> master, origin/master) Update index with CloudEthiX and
Hostname
5fa2461 Initial commit with nginx application
```

```
achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ vim Dockerfile
achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ cat Dockerfile
FROM docker.io/library/nginx:alpine
COPY default.conf /etc/nginx/conf.d/
COPY index.html /usr/share/nginx/html/
-----

achal@LAPTOP-# podman build -t cloudethix-nginx-achal:v1 .

achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ podman images
REPOSITORY                                TAG          IMAGE ID      CREATED
SIZE
localhost/cloudethix-nginx-achal    v1          97ebac177727  29 seconds
ago  64.2 MB
-----

achal@LAPTOP-# kubectl create namespace cloudethix-achal
namespace/cloudethix-achal created
achal@LAPTOP-HPUOTJF1:~/Cloudethix-nginx-achal$ kubectl get namespace
NAME          STATUS    AGE
cloudethix-achal  Active   29s
-----

achal@LAPTOP-# podman tag localhost/achalpadol/cloudethix-nginx-achal:v1 \
docker.io/achalpadol/cloudethix-nginx-achal:v1
-----

achal@LAPTOP-# podman images
REPOSITORY                                TAG          IMAGE ID
CREATED          SIZE
localhost/cloudethix-nginx-achal          v1          97ebac177727
21 minutes ago  64.2 MB
localhost/achalpadol/cloudethix-nginx-achal  v1          97ebac177727
21 minutes ago  64.2 MB
docker.io/achalpadol/cloudethix-nginx-achal  v1          97ebac177727
21 minutes ago  64.2 MB
-----

achal@LAPTOP-# podman push docker.io/achalpadol/cloudethix-nginx-achal:v1
Getting image source signatures
-----

chal@LAPTOP-# vim deployment.yml

achal@LAPTOP-# kubectl apply -f deployment.yml
deployment.apps/cloudethix-nginx created

achal@LAPTOP-# kubectl get pod -n cloudethix-achal
NAME                                READY   STATUS    RESTARTS   AGE
cloudethix-nginx-5d77d6699f-7wkwp  1/1     Running   0          26s
cloudethix-nginx-5d77d6699f-ktvsq  1/1     Running   0          26s
```